

Immunogenicity and Impact on Nasopharyngeal Carriage of a Nonavalent Pneumococcal Conjugate Vaccine

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The safety, immunogenicity, and impact on carriage of a nonvalent pneumococcal vaccine given at ages 6, 10, and 14 weeks were examined in a double-blind, randomized, placebo-controlled trial in 500 infants in Soweto, South Africa. No serious local or systemic side effects were recorded. Significant antibody responses to all pneumococcal serotypes were observed 4 weeks after the third dose. *Haemophilus influenzae* type b polyribosylribitol phosphate (geometric mean titer, 11.62 µg/mL) and diphtheria (1.39 IU/mL) antibodies were significantly higher in children receiving pneumococcal conjugate, compared with placebo recipients (4.58 µg/mL and 0.98 IU/mL, respectively). Nasopharyngeal carriage of vaccine serotypes decreased in vaccinees at age 9 months (18% vs. 36%), whereas carriage of nonvaccine serotypes increased (36% vs. 25%). Carriage of penicillin-resistant pneumococci (21% vs. 41%) and cotrimoxazole-resistant pneumococci (23% vs. 35%) were significantly reduced 9 months after vaccination, compared with controls.

Respiratory tract infections are a major cause of illness and death in the pediatric population in developing countries [1]. It is estimated that 70%–80% of severe pneumonias in Africa are caused by the pneumococcus [2, 3]. Over the past decade there has been an increase from 28% to 45% in the isolation of penicillin-resistant pneumococci causing systemic infections at Chris Hani Baragwanath Hospital in Soweto, South Africa [4].

The impact of conjugate *Haemophilus influenzae* type b (Hib) vaccines on *Haemophilus influenzae* type b invasive disease and carriage has been dramatic [5, 6]. The inability of children to mount significant immune responses to polysaccharide antigens, the prevalence of multiple serotypes of pneumococci causing invasive disease, and the increase in antibiotic resistance have led to the development of polyvalent pneumococcal conjugate vaccines [7, 8]. The immunogenicity of a 7-valent pneumococcal conjugate vaccine containing serotypes 4, 6B, 9V, 14,

18C, 19F, and 23F has recently been reported [9]. In developing countries serotypes 1 and 5 are an important cause of invasive pneumococcal disease [10]; in South Africa, serotype 1 causes 13.5% of invasive pneumococcal disease in children [11].

A randomized, double-blind study of the safety, immunogenicity, and impact on carriage of a 9-valent pneumococcal conjugate vaccine was undertaken. The vaccine contained 2 µg of the carbohydrate of serotypes 1, 4, 5, 9V, 14, 18C, 19F, and 23F carbohydrate and 4 µg of serotype 6B conjugated to mutant diphtheria toxin CRM₁₉₇. These serotypes constitute >85% of pneumococci isolated from the blood or cerebrospinal fluid of pediatric patients admitted to the Chris Hani Baragwanath Hospital in Soweto, South Africa [12].

Vaccines could be a novel way to reduce transmission of resistant strains by reducing carriage in immunized subjects [12]. This is the first report of the safety, immunogenicity, and impact on carriage of a 9-valent pneumococcal conjugate vaccine.

Patients and Methods

Subjects. Infants were recruited from the population presenting for routine immunization at Zola Family Health Centre, a clinic in Soweto, South Africa, after verbal and written informed consent in the preferred language was obtained from the guardians of the subjects.

Immunization. Subjects were individually randomized in a double-blind manner to receive either 9-valent pneumococcal conjugate vaccine or an identical-appearing placebo. The trial vaccine was a lyophilized preparation provided by Wyeth Lederle Vaccines and Pediatrics (WLVP, West Henrietta, NY). Subjects were immunized according to the WHO Expanded Programme on Immunization (EPI) schedule at ages 6, 10, and 14 weeks with HbOC-diphtheria–tetanus toxoids–pertussis (DTP; Tetramune; WLVP), oral polio

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Informed consent was obtained from the guardians of all subjects, and approval for the study was obtained from the University of the Witwatersrand Committee for Studies on Human Subjects and the Medicines Control Council of South Africa.

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Table 1. Immunogenicity of a 9-valent pneumococcal conjugate vaccine at age 18 weeks.

Group	Serotype								
	1	4	5	6B	9V	14	18C	19F	23F
GMC vaccinees ^a	5.82	4.22	7.31	5.60	3.62	3.76	5.68	2.91	2.95
No. tested	122	132	123	132	135	129	132	132	130
GMC controls	0.15	0.06	0.29	0.19	0.14	0.20	0.11	0.25	0.12

NOTE. GMC, geometric mean concentration.

^a Statistically significant compared with controls ($P < .001$).

(Poliorol trivalent; Biocine, Siena, Italy), hepatitis b (Hepacine; Cheil Sugar Organization, Seoul, Korea), and the trial vaccine or placebo. As Hib vaccine is not yet available as part of the public sector vaccination program in South Africa, this vaccine was made available to all the children in the study as a benefit of participation. All children had previously received bacille Calmette-Guérin (BCG) and oral polio vaccine at birth. Injections of 0.5 mL of trial vaccine or placebo, HbOC-DTP, and hepatitis B vaccines were given intramuscularly in the left thigh, right lower thigh, and right upper thigh, respectively.

Monitoring for side effects. Infants were observed for 30 min after immunization. Guardians were requested to record axillary temperatures with digital thermometers supplied by the trialist and to record local and any other reactions that occurred within 72 h after immunization. Telephone contact or home visits were made with the guardians at 24 and 48 h and as required.

Determination of vaccine immunogenicity and effect on pneumococcal carriage. Blood samples and nasopharyngeal swabs were taken at ages 6, 10, 14, and 18 weeks and 9 months. The nasopharyngeal swabs were taken by trained study nurses with a calcium alginate swab [13] and immediately plated onto medium containing 5% horse blood and 5 µg/mL gentamicin. Pneumococci were isolated and identification made by colony morphology and susceptibility to optochin. Serotyping was done by quellung reaction by use of antisera obtained from the Statens Serum Institut (Copenhagen). Antibiotic resistance was initially determined by disc diffusion and confirmed by determination of minimum inhibitory concentrations by using the microdilution method. Results were interpreted according to NCCLS criteria [14]. The serum was separated, stored at -70°C , and shipped to WLVP on dry ice. Standardized ELISAs were used to measure IgG antibody titers to *Streptococcus pneumoniae* [15], Hib [16], and diphtheria and tetanus [17]. All pneumococcal serologic testing was done at WLVP. In addition, human immunodeficiency virus (HIV) ELISA (automated Abbot AxSYM) was performed by the serology department, SAIMR, Johannesburg, on the sera taken at age 9 months.

Data management and statistical analyses. Data management and analysis were performed by using the Epi Info computer software, version 6 (Centers for Disease Control and Prevention, Atlanta, GA) and SAS procedures guide, version 6 (SAS Institute, Cary, NC). Geometric mean concentrations for vaccine serotype-specific IgG antibodies were compared between the vaccine and control groups by use of analysis of covariance on a logarithmic scale. Comparisons of the carriage rates between the treatment groups were made by using Fisher's exact test.

Results

Description of study population. Five hundred infants were randomized into 2 equal groups receiving either vaccine or placebo.

The mean age at recruitment was 45 days (SD, 4.27 days). The mean age at the second visit was 76 days (SD, 9.69 days), at the third vaccination 106 days (SD, 12.9 days), and at 9 months was 282 days (SD, 18.5 days). There was no statistical difference in ages between the 2 groups.

A total of 492 children (98%) completed the 3-dose series and 481 (96%) completed the 9-month follow-up. Nineteen (3.8%) subjects withdrew from the study by 9 months. Reasons for withdrawal included death (6 vaccinees and 4 controls, $P = .5$), religious conviction (1 infant) and loss to follow-up (8 infants). There was no significant difference in the number of deaths or loss to follow-up between the group receiving vaccine and those receiving placebo. Twenty-six children (5.2%) were hospitalized on 30 occasions; the most common reasons for hospitalization were bronchopneumonia (11 cases) and gastroenteritis (5 cases). Of those hospitalized for bronchopneumonia, 2 admissions were for a single subject who had repeated episodes of a pertussis-like syndrome. There was no significant difference in hospitalizations for vaccinees and controls (11 vs. 15, $P = .4$). HIV infection was suspected in 8 (31%) of the 26 children who were hospitalized; 6 (75%) of these 8 children were seropositive.

Vaccine side effects. Local and systemic side effects were monitored after every immunization. Pain was reported more frequently at the site of HbOC-DTP administration than at either the hepatitis B, conjugate pneumococcal vaccine, or placebo sites (6.4% vs. 3.2%, 1.6%, and 4.0%, respectively, $P = .05$). There were no differences in other local reactions between vaccinees and controls. Mothers reported fussiness on the day of immunization in almost 30% of children in both the conjugate and placebo groups, and 15% of the mothers reported giving paracetamol. Other complaints were uncommon. No statistical differences were seen in systemic side effects between the groups, with the exception that controls fed more poorly on the day of immunization (4% vs. 1.2%, $P = .05$). A transient urticarial rash developing >4 h after immunization was reported in 2 children given placebo and in 1 child receiving conjugate vaccine.

Vaccine immunogenicity. The vaccine serotype-specific IgG geometric mean concentrations (GMC) measured at 18 weeks, 4 weeks after the third vaccine dose, are shown in table 1. The vaccine serotype-specific GMCs of the vaccinees ranged from 2.91 to 7.31 µg/mL and were significantly higher than those of control children. Serotype 6B, previously considered a poorly

Table 2. Impact of a 9-valent pneumococcal conjugate vaccine on carriage of *Streptococcus pneumoniae*.

Serotype	No. (%) of children carrying serotype		P
	Vaccinees (n = 242)	Controls (n = 239)	
1 ^a	0	0	—
3	1 (0.4)	2 (0.8)	—
4 ^a	3 (1.2)	5 (2.1)	—
5 ^a	0	0	—
6A	14 (5.8)	11 (4.6)	—
6B ^a	8 (3.3)	28 (11.7)	<.001
7	4 (1.6)	0	.06
8	2 (0.8)	0	—
9A	0	1 (0.4)	—
9L	0	1 (0.4)	—
9N	0	2 (0.8)	—
9V ^a	1 (0.4)	2 (.8)	—
11	3 (1.2)	2 (.8)	—
14 ^a	3 (1.2)	6 (2.5)	—
15	22 (9.1)	9 (3.8)	.017
16	4 (1.6)	1 (0.4)	—
17	4 (1.6)	1 (0.4)	—
18C ^a	2 (0.8)	0	—
19A	7 (2.9)	11 (4.6)	—
19B	1 (0.4)	0	—
19F ^a	19 (7.8)	32 (13.4)	.05
20	2 (0.8)	1 (0.4)	—
21	9 (3.7)	5 (2.1)	—
22	2 (0.8)	2 (0.8)	—
23A	2 (0.8)	0	—
23F ^a	7 (2.9)	14 (5.8)	—
24	1 (0.4)	1 (0.4)	—
29	1 (0.4)	1 (0.4)	—
34	7 (2.9)	6 (2.5)	—
35	1 (0.4)	1 (0.4)	—
All vaccine serotypes	43 (18)	87 (36)	<.001
All nonvaccine serotypes	87 (36)	58 (24)	.007

^a Serotypes in the 9-valent pneumococcal conjugate vaccine.

immunogenic antigen, stimulated a good mean antibody response (5.53 μ g/mL).

Pneumococcal polysaccharide conjugated to CRM₁₉₇ enhanced the antibody response to diphtheria antigen (GMC 1.39 vs. 0.98 IU/mL, $P < .001$) and Hib polyribosylribitol phosphate conjugated to CRM (GMC 11.62 vs. 4.58 μ g/mL, $P = .02$) but not to tetanus (GMC 12.7 vs. 14.24 IU/mL, $P = .64$).

HIV antibody assays were done on 480 of the 490 children surviving at age 9 months. Of these, 427 were negative, 5 indeterminate, and 45 low positive (probably indicating residual maternal antibody); 3 were strongly positive. Pneumococcal antibodies were available from 26 children with low positive or strongly positive HIV serology. This study was not of sufficient statistical power to answer the question of immunogenicity in HIV-infected children. The mean antibody levels for any pneumococcal serotype in children with maternal HIV antibodies were not significantly different from those with HIV negative results. One of the 3 HIV-infected children received vaccine and had antibodies measured. This child mounted a strong response (>5 μ g/mL) to all antigens.

Effect of immunization on pneumococcal carriage. Carriage rates of pneumococci increased with age in both immunization

groups. At age 6 weeks pneumococci were isolated from 64 (26%) of 250 and 74 (30%) of 250 of the vaccinees and controls, respectively. Carriage rates in vaccinees and controls at 10 weeks were 110 (44.2%) of 249 and 109 (43.8%) of 249 and were 115 (46.7%) of 246 and 127 (51.4%) of 247 at 14 weeks. The minimum average length of carriage of the same serotype over the 3-month period of immunization was estimated to be 1.4 months. By age 9 months, 130 (54%) of 242 of the vaccinated children and 145 (61%) of 239 of the control children carried pneumococci. Although no difference in carriage of vaccine and nonvaccine serotypes was detected between the 2 groups at age 18 weeks, at age 9 months the carriage of vaccine-associated serotypes was significantly reduced from 86 (36%) of 239 in control infants to 43 (18%) of 242 in vaccinees (table 2, $P < .001$). The carriage of nonvaccine-associated serotypes increased from 59 (25%) of 239 to 87 (36%) of 242 in vaccinees ($P = .007$). The vaccine serotypes most commonly isolated in both groups were serotypes 6B, 19F, and 23F (table 2). Serotypes 6B and 19F were significantly less common in vaccinated children. Serotype 23F was decreased by 50%, but this difference did not reach statistical significance. The most common nonvaccine serotypes isolated from vaccinees were 15, 6A, and 19A; type 15 was significantly increased in vaccinated children, compared with controls ($P = .017$). Antibiotic-resistant pneumococci were significantly less commonly carried in vaccinated children (table 3). Carriage of penicillin-resistant pneumococci, regardless of serotype, was significantly reduced in vaccinees (21% vs. 41%, $P < .001$). Cotrimoxazole resistance was also reduced from 35% to 23% ($P < .003$) after immunization with the conjugate vaccine.

Discussion

The 9-valent pneumococcal conjugate vaccine appears to be safe when administered in 3 doses to young infants. The vaccine was well tolerated by the infants, with most of the local reactivity localized to the site of DTP injection. There was no

Table 3. Impact of a 9-valent pneumococcal conjugate vaccine on carriage of antibiotic-resistant *Streptococcus pneumoniae*.

Antibiotic resistance ^a	No. (%) of children carrying resistant pneumococci		P
	Vaccinees (n = 130) ^b	Controls (n = 145) ^b	
Penicillin	27 (21)	60 (41)	.0002
Chloramphenicol	2 (2)	5 (3)	—
Tetracycline	14 (11)	13 (9)	—
Erythromycin	8 (6)	6 (4)	—
Clindamycin	7 (5)	4 (3)	—
Rifampicin	2 (2)	1 (1)	—
Cotrimoxazole	30 (23)	51 (35)	.003
Any of the above	59 (45)	90 (62)	.005

^a Resistant to >0.06 μ g/mL penicillin, >4 μ g/mL chloramphenicol, >2.00 μ g/mL tetracycline, >0.25 μ g/mL erythromycin, >0.25 μ g/mL clindamycin, >1.00 μ g/mL rifampicin, >0.5/9.5 μ g/mL cotrimoxazole.

^b n, total number of children carrying a pneumococcus at age 9 months in each vaccination group.

enhancement of systemic toxicity associated with administration of the vaccine compared with placebo. While there was no difference in side effects between the vaccinated and control groups, the overall rates of reactogenicity were low. Our surveillance for adverse reactions was passive and relied on parental assessment. It should be noted that this study, performed on the WHO EPI schedule at ages 6, 10, and 14 weeks in children who have received BCG and oral polio at birth and whole-cell pertussis vaccine, may not be generalizable to infants receiving the standard US vaccine schedule.

Unlike Hib conjugate vaccine, which is generally less immunogenic in younger children and may be less immunogenic in Alaskan native communities [18], the 9-valent pneumococcal conjugate vaccine stimulated higher levels of antibody in our population than those found in older children vaccinated with a 7-valent vaccine in Finland [19] or the United States [9]. In our study, the nasopharyngeal carriage rate of pneumococci was 28% at age 6 weeks. Such early exposure to homologous antigens may prime children in developing countries to more aggressively respond to pneumococcal vaccines.

Although our data are not directly comparable to studies using a 7-valent vaccine in different populations [9, 19], they do suggest that the addition of 2 serotypes to the vaccine is not associated with evidence of hyporesponsiveness. The immunogenicity of the CRM₁₉₇ carrier appears to also enhance responses to the diphtheria component of DTP and responses to the Hib antigens conjugated to CRM₁₉₇, suggesting enhanced carrier priming.

The absolute level of antibody required to protect against invasive pneumococcal disease in infants is unknown. In infant rats, there appear to be differences in levels of serotype-specific antibodies required to confer passive protective immunity [20]. A predictor of immunity to invasive pneumococcal disease after administration of conjugate vaccine is likely to be the induction of memory T cells, as measured by the induction of a booster response to subsequent exposure to pneumococcal carbohydrate antigens. Such a booster response has been observed after initial immunization with a 7-valent pneumococcal vaccine and boosting with a pneumococcal carbohydrate vaccine at age 18 months [21].

The Sowetan infants in this study were colonized with pneumococci at a young age. Early colonization in indigenous populations has been associated with significant morbidity due to otitis media [22]. The 9-valent pneumococcal conjugate vaccine reduced the incidence of carriage of homologous serotypes by >50%. Vaccination may therefore reduce the public health expenditure on the management of antibiotic-resistant otitis media. The carriage of antibiotic-resistant strains in the community has also been correlated with the incidence of antibiotic resistance among invasive pneumococcal isolates in countries such as South Africa [23], Pakistan [24], Israel [25], and Canada [26]. Interruption of carriage of antibiotic resistant strains may, therefore, reduce the incidence of antibiotic-resistant invasive

pneumococcal disease. In the current study, carriage was not reduced at 1 month after vaccination, suggesting that the vaccine does not eliminate strains already present in the nasopharynx but may prevent subsequent acquisition of homologous serotypes. The mechanism by which the vaccine interrupts carriage is unknown but may be due to the high levels of IgG antibody stimulated by the vaccine. In the Gambia, small numbers of children who received 2 or 3 doses of a 5-valent conjugate pneumococcal vaccine had reduced carriage of homologous serotypes, and antigenic replacement with heterologous types was found in 26 children who received all 3 doses of vaccine [27]. Reduction in carriage of homologous types and antibiotic resistant strains, but no significant increase in non-vaccine serotypes, was shown after immunization of toddlers in Israel in the second year of life [28].

The current study was designed to detect reduction and replacement of carriage of specific serotypes. Our data suggest that vaccination significantly reduces carriage of serotypes 19F and 6B. Serotypes 6A and 6B differ only in the linkage of a single sugar in the carbohydrate molecule [29], and it is of some concern that the vaccine did not reduce carriage of serotype 6A. Fifteen percent of the residual carriage of antibiotic-resistant strains in this study was due to the continued carriage of serotype 6A. Our results would seem to correlate with those of Nahm et al. [30], who found that polysaccharide from serotype 6B induces cross-reacting antibodies that have little opsonic activity against serotype 6A. It may, therefore, be important for future generations of vaccine to include this serotype. It is unlikely that pneumococcal vaccine efficacy trials currently under way will have sufficient power to detect the impact of any of the vaccines on invasive disease caused by serotype 6A. The levels of cross protection required to prevent invasive disease may, however, be lower than that required to interrupt transmission of carriage. As increasing levels of antibiotic resistance among pneumococci restrict the serotypes causing invasive disease to those associated with resistance [12], it is likely that the impact of pneumococcal conjugate vaccines will be greatest in those populations that have the highest levels of resistance. Natural populations of pneumococci may switch their capsular genes [31], and the selective pressure of conjugate vaccination may lead to the spread of resistance genes to strains with non-vaccine-associated capsules. As the capsule is the major virulence determinant of the pneumococcus, it is probable that the nonvaccine serotypes will remain less invasive. The introduction of conjugate Hib vaccines has not increased the incidence of *Haemophilus* disease due to serotypes a and c-f. Of some concern is the possibility that the incidence of invasive disease due to virulent pneumococcal serotypes not included in the vaccine, such as serotype 15, may increase as the carriage of that serotype appears to be enhanced or unmasked by elimination of vaccine serotypes. Excluding vaccine serotypes, serotype 15 is one of the most common serotypes isolated from blood and cerebrospinal fluid in South African children [11].

The poor sensitivity of current methods to detect carriage of multiple strains of pneumococci raises questions as to the mechanism of serotype substitution seen in this study. It is possible that the nonvaccine serotypes found in excess in the vaccinated population were truly exogenously acquired, but it is also possible that these strains represent subdominant clones of carried bacteria that are not detected by a nasopharyngeal culture; for example, they may be present at concentrations of >2 log less than the dominant strain and, therefore, would be rarely detected by serotyping only 5 colonies. Although multiple passage of sputum through mice has been shown to detect up to 5 different serotypes of pneumococci in a single sample [32], less expensive and cumbersome approaches for the detection of multiple serotype carriage are required.

The immunogenicity and impact on carriage of pneumococcal 9-valent conjugate vaccine in HIV-infected infants is unknown. It is important that these issues be investigated in this group of children, as failure of interruption of pneumococcal carriage in HIV-infected children may allow a reservoir of carriage of vaccine serotypes and antibiotic-resistant strains in the community. These bacteria could then place unimmunized young infants at risk and reduce the herd immunity of the vaccine. The efficacy of this vaccine in HIV-infected children is currently under study.

In conclusion, this study suggests that pneumococcal conjugate vaccines may reduce the burden of pneumococcal disease and antimicrobial resistance in pneumococcal strains. Vaccination interrupts carriage of homologous serotypes, which may be indirect evidence of potential efficacy against invasive disease. Direct demonstration of protection and the impact, if any, of the apparently enhanced carriage of nonvaccine serotypes on invasive disease and severe pneumonia is currently under investigation.

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